

Commodore VIC-20 Emulator User Guide

STARTUP OPTIONS AND COMMAND CONSOLE REFERENCE

IGNACIO CEA FORNIÉS

Contents

CONTENTS 1

/O 6

/P 6

/H 6

/F 7

/K 7

/C 7

/L 8

/A 8

/D 8

/S 9

/I 9

/R 9

/G 10

/N 10

/W 10

/B 11

/H 12

/F 12

/K 12

/C 13

/L 13

/A.....	13
/D.....	14
/S.....	14
/I.....	14
/R.....	15
/P.....	15
/G.....	15
/N.....	16
/W.....	16
/B.....	16
3.1 STANDARD FRAMEWORK COMMANDS	18
HELP	18
AUTHOR	18
STATUS	18
PC.....	19
REGISTERS	19
CHANGereg	19
STACK.....	20
CPUSTATUS.....	20
CPUSSTATUS.....	20
CPUSTATE	21
CPUINFO.....	21
MEMORY	21
MEMORYDUMP.....	22
MEMORYSTR	22
SETMEMORY.....	22
MEMORYANALYSIS	23
STOP	23
RUN	24
SETPC	24
NEXT	24
NEXTSTACK	25
SHOWNEXT	25
INST	25
BREAKS.....	26
SETBREAK	26
REMOVEBREAK.....	26
REMOVEBREAKS.....	27
SPEED	27
LOADBIN	27
SAVEBIN.....	28

DEBUGON	28
DEBUGOFF	28
IDEBUGON	29
IDEBUGOFF	29
RESTART.....	29
DEVICES.....	30
PERIPHERALS	30
PERCOMMAND	30
ASSIGNJOY.....	31
CLOCKFACTOR.....	31
SOUNDON.....	31
SOUNDOFF.....	32
INTERRUPTS	32
INTSET	32
CHIPS	33
CRTON	33
CRTOFF.....	33
DATASETTE.....	34
GRIDON	34
GRIDOFF	34
PICTURE.....	35
SETHOOK	35
REMOVEHOOK.....	35
HOOKS.....	36
HOOKSHELP.....	36
CMOVEPARAMS.....	36

3.2 INHERITED COMMODORE COMMANDS.....37

VICII.....	37
VICIIEVENTS	37
SID	37
SIDW.....	38
VICI.....	38
TED	39
TEDEVENTS	39
VIA	39
CIA.....	40

3.3 VIC20-SPECIFIC COMMANDS.....40

VIA1	40
VIA2	40
SCREENDUMP	41
COLORDUMP.....	41
CHARSDRAW.....	41
LIGHTPEN.....	42

3.4 LOCAL-CONSOLE-ONLY COMMANDS.....42

LOADPRG	42
LOADBINARY	42
LOADBLOCKS	43
DECOMPILE.....	43

POSSIBLEPERS	43
CONNECTPER.....	44
DISCONNECTPERS	44
LOADPERDATA	44
EMPTYPERDATA.....	45
SAVEPERDATA	45
CLEARPERDATA.....	45
 APPENDIX A. COMMAND AVAILABILITY SUMMARY	 47
 APPENDIX B. SOURCE-OF-TRUTH MAINTENANCE RULE	 47
 ILLUSTRATIONS	 48
 EQUATIONS	 49

Foreword

This guide describes the startup interface and the command interface implemented by the two Commodore VIC-20 executables in EMULATORS: VIC20Emulator (without a local command console) and VIC20EmulatorC (with a local command console). It is based on the executable entry points, the VIC20Emulator inheritance chain, LocalConsole, and the complete VIC-20 -> COMMODORE -> StandardCommandBuilder responsibility chain.

Command and parameter names are shown exactly as accepted by the current code. Startup options use /XVALUE with no space between the one-letter option and its value. Command names are matched as written by the builders; use uppercase names for reliable operation. Positional command parameters are separated by spaces; examples use paths without embedded spaces.

The non-console executable does not expose an interactive prompt. When started with /o it creates a remote communication system using the same VIC-20, Commodore and standard command builders. Commands marked Local console only are intercepted by LocalConsole and therefore cannot be sent to the non-console executable.

IGNACIO CEA FORNIES

1. Startup Parameters: Version Without a Local Console

Executable: VIC20Emulator.exe. It accepts the framework and VIC-20 options below, plus /o and the communication port option. Option letters are case-sensitive because the command-line parser stores the character exactly as supplied.

/o

Meaning: Controls creation of the remote command communication system.

General form:

```
/o[YES|NO]
```

- YES or an empty value: open remote communications.
- NO (and any other non-YES non-empty value): do not open it.
- Omitted: remote communications disabled.

Examples:

```
VIC20Emulator.exe /o  
VIC20Emulator.exe /oYES /p60000
```

Consequence: When enabled, remote clients can execute VIC-20, Commodore and standard builder commands. LocalConsole-only commands remain unavailable.

/p

Meaning: Selects the TCP listening port used when /o enables remote communications.

General form:

```
/pPORT
```

- PORT: decimal port number; default 60000 when /p is omitted.

Examples:

```
VIC20Emulator.exe /o /p60000  
VIC20Emulator.exe /o /p61000
```

Consequence: The communication system attempts to listen on the converted port.

Important: Current-code collision: the inherited Emulator also interprets the same /p value as a comma-separated peripheral ID list. A normal port value can therefore trigger an unintended peripheral connection attempt. Avoid combining startup peripheral attachment with the non-console executable until the option letters are separated in code.

/h

Meaning: Prints the executable help and prevents the emulation from starting.

General form:

```
/h
```

- No value.

Examples:

```
VIC20Emulator.exe /h
```

Consequence: Initialization returns failure after printing the help; the current main program therefore exits with code 1.

/f

Meaning: Loads a binary file whose header contains the destination address.

General form:

```
/fFILE
```

- FILE: path to the header-addressed binary file; attach it directly to /f.

Examples:

```
VIC20Emulator.exe /fprogram.prg
VIC20Emulator.exe /fgames\demo.prg /a$080d
```

Consequence: The file is loaded before execution. An unreadable or invalid file aborts initialization.

/k

Meaning: Loads a block-structured memory file.

General form:

```
/kFILE
```

- FILE: path to a block file understood by the framework.

Examples:

```
VIC20Emulator.exe /ksnapshot.blocks
```

Consequence: Every encoded block is placed at its stored address. A load error aborts initialization.

/c

Meaning: Parses, assembles and loads an assembler source file.

General form:

```
/cFILE
```

- FILE: assembler source path.

Examples:

```
VIC20Emulator.exe /cstartup.asm /s
```

Consequence: The assembled bytes are installed before the main loop. Syntax or load errors are logged and initialization stops.

/l

Meaning: Selects the framework diagnostic level.

General form:

```
/lLEVEL
```

- 0: no diagnostics.
- 1: errors (default).
- 2: dump at exit.
- 3: errors and warnings.
- 4: all regular diagnostics.
- 5: internal trace diagnostics.

Examples:

```
VIC20Emulator.exe /l0  
VIC20Emulator.exe /l4
```

Consequence: Higher values produce progressively more diagnostic output and may reduce performance.

/a

Meaning: Overrides the address at which execution begins.

General form:

```
/aADDRESS
```

- ADDRESS: VIC-20 address in the framework number syntax, for example decimal or hexadecimal with \$.

Examples:

```
VIC20Emulator.exe /a$080d  
VIC20Emulator.exe /fprogram.prg /a2061
```

Consequence: A non-zero, valid address replaces the normal reset/program entry address.

/d

Meaning: Creates startup stop points at one or more addresses.

General form:

```
/dADDRESS1,ADDRESS2,...
```

- ADDRESSn: comma-separated address. Each entry accepts the framework number syntax.

Examples:

```
VIC20Emulator.exe /d$080d  
VIC20Emulator.exe /d$080d,$ffd2 /s
```

Consequence: Execution stops when the CPU reaches any listed address, allowing later inspection or continuation.

/s

Meaning: Controls whether the emulated computer starts stopped.

General form:

```
/s[YES|NO]
```

- YES or an empty value: start stopped.
- NO (and any non-YES non-empty value): start running.
- Omitted: start running.

Examples:

```
VIC20Emulator.exe /s  
VIC20Emulator.exe /sNO
```

Consequence: When enabled, the machine is initialized but the CPU does not run until a RUN/NEXT command or the equivalent UI action is issued.

/i

Meaning: Selects the VIC-20 language/character-ROM variant.

General form:

```
/iLANG
```

- ENG: English (default).
- JAP: Japanese.
- SWE: Swedish.
- DKA: Danish.

Examples:

```
VIC20Emulator.exe /iENG
```

Consequence: The selected language is used when constructing the VIC-20 model and its ROM-dependent presentation.

/r

Meaning: Controls sound at startup.

General form:

```
/r[YES|NO]
```

- YES or an empty value: sound enabled.
- NO (and any non-YES non-empty value): sound disabled.
- Omitted: sound enabled.

Examples:

```
VIC20Emulator.exe /r  
VIC20Emulator.exe /rNO
```

Consequence: The audio subsystem begins enabled or disabled. The state can later be changed with SOUNDON/SOUNDOFF.

/g

Meaning: Controls the CRT post-processing effect at startup.

General form:

```
/g[YES|NO]
```

- YES or an empty value: CRT effect enabled.
- NO (and any non-YES non-empty value): disabled.
- Omitted: disabled.

Examples:

```
VIC20Emulator.exe /g  
VIC20Emulator.exe /gNO
```

Consequence: The screen starts with or without CRT emulation. CRTON and CROFF can change it later.

/n

Meaning: Selects NTSC timing instead of PAL.

General form:

```
/n
```

- No value; presence selects NTSC.

Examples:

```
VIC20Emulator.exe /n
```

Consequence: The emulator creates an NTSC VIC-20. Without /n it creates the PAL model.

/w

Meaning: Selects the VIC-20 memory-expansion configuration used at startup.

General form:

```
/w[CONF]
```

- 0 or omitted: unexpanded VIC-20 (default).
- 1: +3 KB expansion.
- 2: +8 KB expansion.
- 3: +16 KB expansion (+8 KB +8 KB).
- 4: +24 KB expansion (+8 KB +8 KB +8 KB).
- 5: +32 KB expansion (+8 KB +8 KB +8 KB +8 KB).
- 6: +11 KB expansion (+3 KB +8 KB).
- 7: +19 KB expansion (+3 KB +8 KB +8 KB).
- 8: +27 KB expansion (+3 KB +8 KB +8 KB +8 KB).
- 9 or any greater numeric value: +35 KB expansion (+3 KB +8 KB +8 KB +8 KB +8 KB); values above 9 are clamped to 9.

Examples:

```
VIC20Emulator.exe /w0  
VIC20Emulator.exe /w3
```

```
VIC20Emulator.exe /w9
```

Consequence: The selected RAM expansion is installed before the VIC-20 is initialized, changing its visible memory map and the software configurations it can run.

/b

Meaning: Draws the VIC-20 screen grid/border using the requested color.

General form:

```
/bCOLOR
```

- COLOR: numeric color value. If /b is present without a valid number, conversion yields 0.

Examples:

```
VIC20Emulator.exe /b1  
VIC20Emulator.exe /b14 /s
```

Consequence: The screen grid is enabled at startup with the converted color value.

2. Startup Parameters: Version With a Local Console

Executable: VIC20EmulatorC.exe. Its main program delegates the complete command line to VIC20::Emulator, so all inherited framework options and the VIC20-specific options are active even though the short main-level help banner only advertises /h. It has no /o listener option; commands are entered in its local console.

/h

Meaning: Prints the executable help and prevents the emulation from starting.

General form:

```
/h
```

- No value.

Examples:

```
VIC20Emulator.exe /h
```

Consequence: Initialization returns failure after printing the help; the current main program therefore exits with code 1.

/f

Meaning: Loads a binary file whose header contains the destination address.

General form:

```
/fFILE
```

- FILE: path to the header-addressed binary file; attach it directly to /f.

Examples:

```
VIC20Emulator.exe /fprogram.prg  
VIC20Emulator.exe /fgames\demo.prg /a$080d
```

Consequence: The file is loaded before execution. An unreadable or invalid file aborts initialization.

/k

Meaning: Loads a block-structured memory file.

General form:

```
/kFILE
```

- FILE: path to a block file understood by the framework.

Examples:

```
VIC20Emulator.exe /ksnapshot.blocks
```

Consequence: Every encoded block is placed at its stored address. A load error aborts initialization.

/c

Meaning: Parses, assembles and loads an assembler source file.

General form:

```
/cFILE
```

- FILE: assembler source path.

Examples:

```
VIC20Emulator.exe /cstartup.asm /s
```

Consequence: The assembled bytes are installed before the main loop. Syntax or load errors are logged and initialization stops.

/l

Meaning: Selects the framework diagnostic level.

General form:

```
/lLEVEL
```

- 0: no diagnostics.
- 1: errors (default).
- 2: dump at exit.
- 3: errors and warnings.
- 4: all regular diagnostics.
- 5: internal trace diagnostics.

Examples:

```
VIC20Emulator.exe /l0  
VIC20Emulator.exe /l4
```

Consequence: Higher values produce progressively more diagnostic output and may reduce performance.

/a

Meaning: Overrides the address at which execution begins.

General form:

```
/aADDRESS
```

- ADDRESS: VIC-20 address in the framework number syntax, for example decimal or hexadecimal with \$.

Examples:

```
VIC20Emulator.exe /a$080d  
VIC20Emulator.exe /fprogram.prg /a2061
```

Consequence: A non-zero, valid address replaces the normal reset/program entry address.

/d

Meaning: Creates startup stop points at one or more addresses.

General form:

```
/dADDRESS1,ADDRESS2,...
```

- ADDRESSn: comma-separated address. Each entry accepts the framework number syntax.

Examples:

```
VIC20Emulator.exe /d$080d  
VIC20Emulator.exe /d$080d,$ffd2 /s
```

Consequence: Execution stops when the CPU reaches any listed address, allowing later inspection or continuation.

/s

Meaning: Controls whether the emulated computer starts stopped.

General form:

```
/s[YES|NO]
```

- YES or an empty value: start stopped.
- NO (and any non-YES non-empty value): start running.
- Omitted: start running.

Examples:

```
VIC20Emulator.exe /s  
VIC20Emulator.exe /sNO
```

Consequence: When enabled, the machine is initialized but the CPU does not run until a RUN/NEXT command or the equivalent UI action is issued.

/i

Meaning: Selects the VIC-20 language/character-ROM variant.

General form:

```
/iLANG
```

- ENG: English (default).
- JAP: Japanese.
- SWE: Swedish.
- DKA: Danish.

Examples:

```
VIC20Emulator.exe /iENG
```

Consequence: The selected language is used when constructing the VIC-20 model and its ROM-dependent presentation.

/r

Meaning: Controls sound at startup.

General form:

```
/r[YES|NO]
```

- YES or an empty value: sound enabled.
- NO (and any non-YES non-empty value): sound disabled.
- Omitted: sound enabled.

Examples:

```
VIC20Emulator.exe /r  
VIC20Emulator.exe /rNO
```

Consequence: The audio subsystem begins enabled or disabled. The state can later be changed with SOUNDON/SOUNDOFF.

/p

Meaning: Connects framework peripherals during initialization.

General form:

```
/pID1, ID2, ...
```

- 10: typewriter input.
- 100: 1530 datasette.
- 101: injected/fast 1530 datasette.
- 203: cartridge.
- 205 or 206: serial printer instances.
- 215, 216, 217 or 218: 1541 disk-drive instances.

Examples:

```
VIC20EmulatorC.exe /p100  
VIC20EmulatorC.exe /p100,101
```

Consequence: Each recognized ID is built and connected before execution; unsupported IDs are not connected.

Important: Use POSSIBLEPERS in the local console to obtain the IDs supported by the current build.

/g

Meaning: Controls the CRT post-processing effect at startup.

General form:

```
/g[YES|NO]
```

- YES or an empty value: CRT effect enabled.
- NO (and any non-YES non-empty value): disabled.
- Omitted: disabled.

Examples:

```
VIC20Emulator.exe /g
```



```
VIC20Emulator.exe /gNO
```

Consequence: The screen starts with or without CRT emulation. CRTON and CRTOFF can change it later.

/n

Meaning: Selects NTSC timing instead of PAL.

General form:

```
/n
```

- No value; presence selects NTSC.

Examples:

```
VIC20Emulator.exe /n
```

Consequence: The emulator creates an NTSC VIC-20. Without /n it creates the PAL model.

/w

Meaning: Selects the VIC-20 memory-expansion configuration used at startup.

General form:

```
/w[CONF]
```

- 0 or omitted: unexpanded VIC-20 (default).
- 1: +3 KB expansion.
- 2: +8 KB expansion.
- 3: +16 KB expansion (+8 KB +8 KB).
- 4: +24 KB expansion (+8 KB +8 KB +8 KB).
- 5: +32 KB expansion (+8 KB +8 KB +8 KB +8 KB).
- 6: +11 KB expansion (+3 KB +8 KB).
- 7: +19 KB expansion (+3 KB +8 KB +8 KB).
- 8: +27 KB expansion (+3 KB +8 KB +8 KB +8 KB).
- 9 or any greater numeric value: +35 KB expansion (+3 KB +8 KB +8 KB +8 KB +8 KB); values above 9 are clamped to 9.

Examples:

```
VIC20EmulatorC.exe /w0  
VIC20EmulatorC.exe /w3  
VIC20EmulatorC.exe /w9
```

Consequence: The selected RAM expansion is installed before the VIC-20 is initialized, changing its visible memory map and the software configurations it can run.

/b

Meaning: Draws the VIC-20 screen grid/border using the requested color.

General form:

```
/bCOLOR
```

- COLOR: numeric color value. If /b is present without a valid number, conversion yields 0.

Examples:

```
VIC20Emulator.exe /b1  
VIC20Emulator.exe /b14 /s
```

Consequence: The screen grid is enabled at startup with the converted color value.

IGNACIO CEA FORNIÉS

3. Command Console Reference

The local console checks its eleven emulator-level commands first. All remaining text is delegated through VIC20::CommandBuilder, then COMMODORE::CommandBuilder, and finally MCHEmul::StandardCommandBuilder. The remote /o channel in VIC20Emulator.exe uses only that three-builder chain. The availability line in each entry makes this boundary explicit.

3.1 Standard Framework Commands

HELP

Availability: Local console and remote /o channel.

Accepted aliases: CHELP.

Meaning: Displays the VIC-20 console help assembled from the VIC-20, Commodore, local-console and standard help files.

General syntax:

```
HELP
```

Parameters:

- No parameters.

Examples:

```
HELP
```

Result and consequences: Displays the VIC-20 console help assembled from the VIC-20, Commodore, local-console and standard help files.

AUTHOR

Availability: Local console and remote /o channel.

Accepted aliases: CAUTHOR.

Meaning: Returns framework author information.

General syntax:

```
AUTHOR
```

Parameters:

- No parameters.

Examples:

```
AUTHOR
```

Result and consequences: Returns framework author information.

STATUS

Availability: Local console and remote /o channel.

Accepted aliases: CSTATUS.

Meaning: Shows the 6502 status register.

General syntax:

```
STATUS
```

Parameters:

- No parameters.

Examples:

```
STATUS
```

Result and consequences: Shows the 6502 status register.

PC

Availability: Local console and remote /o channel.

Accepted aliases: CPC.

Meaning: Shows the current program counter.

General syntax:

```
PC
```

Parameters:

- No parameters.

Examples:

```
PC
```

Result and consequences: Shows the current program counter.

REGISTERS

Availability: Local console and remote /o channel.

Accepted aliases: CREGISTERS.

Meaning: Shows all CPU registers.

General syntax:

```
REGISTERS
```

Parameters:

- No parameters.

Examples:

```
REGISTERS
```

Result and consequences: Shows all CPU registers.

CHANGereg

Availability: Local console and remote /o channel.

Accepted aliases: CCHANGereg.

Meaning: Changes one CPU internal register.

General syntax:

```
CHANGereg REGNAME VALUE
```

Parameters:

- REGNAME: registered 6502 register name.
- VALUE: new decimal, z-prefixed binary, o-prefixed octal or \$-prefixed hexadecimal value.

Examples:

```
CHANGereg A $41
```

Result and consequences: The register changes only if it exists and accepts the supplied byte width; otherwise an error is returned.

STACK

Availability: Local console and remote /o channel.

Accepted aliases: CSTACK.

Meaning: Shows stack state.

General syntax:

```
STACK [ALL]
```

Parameters:

- ALL: optional; includes stack contents in addition to basic state.

Examples:

```
STACK  
STACK ALL
```

Result and consequences: Shows stack state.

CPUSTATUS

Availability: Local console and remote /o channel.

Accepted aliases: CCPUSTATUS.

Meaning: Shows registers, status, program counter and stack.

General syntax:

```
CPUSTATUS
```

Parameters:

- No parameters.

Examples:

```
CPUSTATUS
```

Result and consequences: Shows registers, status, program counter and stack.

CPUSSTATUS

Availability: Local console and remote /o channel.

Accepted aliases: CCPUSSTATUS.

Meaning: Shows the compact CPU status without stack detail.

General syntax:

```
CPUSSTATUS
```

Parameters:

- No parameters.

Examples:

```
CPUSSTATUS
```

Result and consequences: Shows the compact CPU status without stack detail.

CPUSTATE

Availability: Local console and remote /o channel.

Accepted aliases: CCPUSTATE.

Meaning: Shows the CPU execution state and state-specific data.

General syntax:

```
CPUSTATE
```

Parameters:

- No parameters.

Examples:

```
CPUSTATE
```

Result and consequences: Shows the CPU execution state and state-specific data.

CPUINFO

Availability: Local console and remote /o channel.

Accepted aliases: CCPUINFO.

Meaning: Returns the complete CPU information structure.

General syntax:

```
CPUINFO
```

Parameters:

- No parameters.

Examples:

```
CPUINFO
```

Result and consequences: Returns the complete CPU information structure.

MEMORY

Availability: Local console and remote /o channel.

Accepted aliases: CMEMORY.

Meaning: Reads the active CPU-visible memory range.

General syntax:

```
MEMORY INITIAL [FINAL]
```

Parameters:

- INITIAL: first address.
- FINAL: optional last address; with no final address only the initial location is reported.

Examples:

```
MEMORY $0400  
MEMORY $0400 $0427
```

Result and consequences: Reads the active CPU-visible memory range.

MEMORYDUMP

Availability: Local console and remote /o channel.

Accepted aliases: CMEMORYDUMP.

Meaning: Dumps every memory view between two addresses, including inactive RAM/ROM views.

General syntax:

```
MEMORYDUMP INITIAL FINAL
```

Parameters:

- INITIAL: first address.
- FINAL: last address.

Examples:

```
MEMORYDUMP $a000 $a03f
```

Result and consequences: Dumps every memory view between two addresses, including inactive RAM/ROM views.

MEMORYSTR

Availability: Local console and remote /o channel.

Accepted aliases: CMEMORYSTR.

Meaning: Shows the configured memory structure.

General syntax:

```
MEMORYSTR [ALL]
```

Parameters:

- ALL: optional; includes inactive regions. Without it only active regions are shown.

Examples:

```
MEMORYSTR  
MEMORYSTR ALL
```

Result and consequences: Shows the configured memory structure.

SETMEMORY

Availability: Local console and remote /o channel.

Accepted aliases: CSETMEMORY.

Meaning: Writes a value to one address or repeatedly across a range.

General syntax:

```
SETMEMORY INITIAL [FINAL] VALUE
```

Parameters:

- INITIAL: first address.
- FINAL: optional last address.
- VALUE: byte sequence in framework numeric syntax; multi-byte values advance by their own length.

Examples:

```
SETMEMORY $0400 $41  
SETMEMORY $0400 $0427 $20
```

Result and consequences: The active memory view is modified immediately; unsafe writes can corrupt state or crash the emulated program.

MEMORYANALYSIS

Availability: Local console and remote /o channel.

Accepted aliases: CMEMORYANALYSIS.

Meaning: Analyzes the current memory configuration.

General syntax:

```
MEMORYANALYSIS
```

Parameters:

- No parameters.

Examples:

```
MEMORYANALYSIS
```

Result and consequences: Analyzes the current memory configuration.

STOP

Availability: Local console and remote /o channel.

Accepted aliases: CSTOP.

Meaning: Stops CPU execution.

General syntax:

```
STOP
```

Parameters:

- No parameters.

Examples:

```
STOP
```

Result and consequences: Stops CPU execution.

RUN

Availability: Local console and remote /o channel.

Accepted aliases: R, CRUN.

Meaning: Starts or resumes CPU execution; alias R.

General syntax:

```
RUN [ADDRESS]
```

Parameters:

- ADDRESS: optional new program counter before running.

Examples:

```
RUN  
R $080d
```

Result and consequences: The CPU resumes immediately, optionally after changing PC.

SETPC

Availability: Local console and remote /o channel.

Accepted aliases: CSETPC.

Meaning: Sets the program counter without starting execution.

General syntax:

```
SETPC ADDRESS
```

Parameters:

- ADDRESS: new CPU address.

Examples:

```
SETPC $080d
```

Result and consequences: Subsequent execution continues from the new address; an unsafe address can break the program.

NEXT

Availability: Local console and remote /o channel.

Accepted aliases: N, CNEXT.

Meaning: Executes the next instruction; alias N.

General syntax:

```
NEXT
```

Parameters:

- No parameters.

Examples:

```
NEXT  
N
```

Result and consequences: Executes the next instruction; alias N.

NEXTSTACK

Availability: Local console and remote /o channel.

Accepted aliases: NS, CNEXTSTACK.

Meaning: Runs until the stack pointer returns to its initial value; alias NS.

General syntax:

```
NEXTSTACK
```

Parameters:

- No parameters.

Examples:

```
NS
```

Result and consequences: Useful for stepping over a routine, although behavior depends on the routine's stack discipline.

SHOWNEXT

Availability: Local console and remote /o channel.

Accepted aliases: SN, CSHOWNEXT.

Meaning: Disassembles upcoming instruction(s); alias SN.

General syntax:

```
SHOWNEXT [NUMBER]
```

Parameters:

- NUMBER: optional count; one instruction by default.

Examples:

```
SHOWNEXT  
SN 8
```

Result and consequences: Disassembles upcoming instruction(s); alias SN.

INST

Availability: Local console and remote /o channel.

Accepted aliases: CINST.

Meaning: Shows the last instruction executed.

General syntax:

```
INST
```

Parameters:

- No parameters.

Examples:

```
INST
```

Result and consequences: Shows the last instruction executed.

BREAKS

Availability: Local console and remote /o channel.

Accepted aliases: CBREAKS.

Meaning: Lists all stop-point addresses.

General syntax:

```
BREAKS
```

Parameters:

- No parameters.

Examples:

```
BREAKS
```

Result and consequences: Lists all stop-point addresses.

SETBREAK

Availability: Local console and remote /o channel.

Accepted aliases: CSETBREAK.

Meaning: Adds one or more breakpoints.

General syntax:

```
SETBREAK ADDRESS1 [ADDRESS2 ...]
```

Parameters:

- ADDRESSn: one or more addresses in framework numeric syntax.

Examples:

```
SETBREAK $080d $ffd2
```

Result and consequences: Adds one or more breakpoints.

REMOVEBREAK

Availability: Local console and remote /o channel.

Accepted aliases: CREMOVEBREAK.

Meaning: Removes selected breakpoints.

General syntax:

```
REMOVEBREAK ADDRESS1 [ADDRESS2 ...]
```

Parameters:

- ADDRESSn: one or more existing breakpoint addresses.

Examples:

```
REMOVEBREAK $ffd2
```

Result and consequences: Removes selected breakpoints.

REMOVEBREAKS

Availability: Local console and remote /o channel.

Accepted aliases: CREMOVEBREAKS.

Meaning: Removes every breakpoint.

General syntax:

```
REMOVEBREAKS
```

Parameters:

- No parameters.

Examples:

```
REMOVEBREAKS
```

Result and consequences: Removes every breakpoint.

SPEED

Availability: Local console and remote /o channel.

Accepted aliases: CSPEED.

Meaning: Shows measured CPU and screen-I/O execution speed.

General syntax:

```
SPEED
```

Parameters:

- No parameters.

Examples:

```
SPEED
```

Result and consequences: Shows measured CPU and screen-I/O execution speed.

LOADBIN

Availability: Local console and remote /o channel.

Accepted aliases: CLOADBIN.

Meaning: Loads a raw binary file at an explicit address.

General syntax:

```
LOADBIN FILE ADDRESS
```

Parameters:

- FILE: raw binary path.
- ADDRESS: first destination address.

Examples:

```
LOADBIN charset.bin $2000
```

Result and consequences: Bytes are copied into memory without an embedded load-address header.

SAVEBIN

Availability: Local console and remote /o channel.

Accepted aliases: CSAVEBIN.

Meaning: Saves a memory range as raw binary.

General syntax:

```
SAVEBIN FILE SIZE ADDRESS
```

Parameters:

- FILE: output path.
- SIZE: byte count.
- ADDRESS: first source address.

Examples:

```
SAVEBIN screen.bin 1000 $0400
```

Result and consequences: Saves a memory range as raw binary.

DEBUGON

Availability: Local console and remote /o channel.

Accepted aliases: CACTIVATEDEEPDEBUG.

Meaning: Enables deep execution debugging.

General syntax:

```
DEBUGON FILE CPU [APPEND] [FILTERS...]
```

Parameters:

- FILE: output log.
- CPU: YES includes CPU execution; NO omits it.
- APPEND: optional YES appends, NO overwrites (default).
- D:START-END or ALLDIR: address filter.
- C:IDs or ALLCHIP: chip filter.
- I:IDs or ALLIO: device filter.
- M:IDs or ALLMEM: memory filter.

Examples:

```
DEBUGON vic20.log YES NO D:$0800-$0fff ALLCHIP ALLIO ALLMEM
```

Result and consequences: Tracing begins with the selected sources and can substantially slow emulation. It cannot coexist with interrupt-only debugging.

DEBUGOFF

Availability: Local console and remote /o channel.

Accepted aliases: CDESACTIVATEDEEPDEBUG.

Meaning: Stops deep execution debugging.

General syntax:

```
DEBUGOFF
```

Parameters:

- No parameters.

Examples:

```
DEBUGOFF
```

Result and consequences: Stops deep execution debugging.

IDEBUGON

Availability: Local console and remote /o channel.

Accepted aliases: CIDEBUGON.

Meaning: Enables interrupt-only debugging.

General syntax:

```
IDEBUGON FILE [APPEND]
```

Parameters:

- FILE: output log.
- APPEND: optional YES appends; NO overwrites (default).

Examples:

```
IDEBUGON interrupts.log NO
```

Result and consequences: Interrupt tracing begins and cannot coexist with deep debugging.

IDEBUGOFF

Availability: Local console and remote /o channel.

Accepted aliases: CIDEBUGOFF.

Meaning: Stops interrupt-only debugging.

General syntax:

```
IDEBUGOFF
```

Parameters:

- No parameters.

Examples:

```
IDEBUGOFF
```

Result and consequences: Stops interrupt-only debugging.

RESTART

Availability: Local console and remote /o channel.

Accepted aliases: CRESTART.

Meaning: Requests a restart and exits the current computer loop so it can be rebuilt.

General syntax:

```
RESTART STOPPED [LEVEL]
```

Parameters:

- STOPPED: YES restarts stopped; NO restarts running.

- LEVEL: optional 0 CPU-only/mild restart (default), 1 computer restart preserving memory, greater than 1 full restart.

Examples:

```
RESTART YES 0
RESTART NO 2
```

Result and consequences: Requests a restart and exits the current computer loop so it can be rebuilt.

DEVICES

Availability: Local console and remote /o channel.

Accepted aliases: CDEVICES.

Meaning: Shows all or selected I/O devices.

General syntax:

```
DEVICES [ID1 ID2 ...]
```

Parameters:

- IDn: optional device identifiers; with none, all devices are returned.

Examples:

```
DEVICES
DEVICES 1 2
```

Result and consequences: Shows all or selected I/O devices.

PERIPHERALS

Availability: Local console and remote /o channel.

Accepted aliases: CPERIPHERALS.

Meaning: Shows all or selected connected peripherals.

General syntax:

```
PERIPHERALS [ID1 ID2 ...]
```

Parameters:

- IDn: optional peripheral identifiers; with none, all are returned.

Examples:

```
PERIPHERALS
PERIPHERALS 100
```

Result and consequences: Shows all or selected connected peripherals.

PERCOMMAND

Availability: Local console and remote /o channel.

Accepted aliases: CPERCMD.

Meaning: Sends a device-specific command to a connected peripheral.

General syntax:

```
PERCOMMAND PERID CMDID [ATTRS...]
```

Parameters:

- PERID: connected peripheral ID.
- CMDID: command ID implemented by that peripheral.
- ATTRS: optional positional attributes forwarded unchanged.

Examples:

```
PERCOMMAND 100 0
```

Result and consequences: The peripheral executes the request if both IDs and its parameter contract are valid; otherwise the answer reports an error.

ASSIGNJOY

Availability: Local console and remote /o channel.

Accepted aliases: CASSIGNJ.

Meaning: Remaps a joystick identifier.

General syntax:

```
ASSIGNJOY ID NEWID
```

Parameters:

- ID: current joystick ID.
- NEWID: required target ID; using the same ID removes an existing assignment.

Examples:

```
ASSIGNJOY 0 1
```

Result and consequences: Remaps a joystick identifier.

CLOCKFACTOR

Availability: Local console and remote /o channel.

Accepted aliases: CCLOCKFACTOR.

Meaning: Changes the CPU clock multiplier.

General syntax:

```
CLOCKFACTOR VALUE
```

Parameters:

- VALUE: numeric factor; 1 means normal speed.

Examples:

```
CLOCKFACTOR 2
```

Result and consequences: The emulation pacing target changes; values above one request faster execution.

SOUNDON

Availability: Local console and remote /o channel.

Accepted aliases: CSOUNDON.

Meaning: Enables audio output.

General syntax:

```
SOUNDON
```

Parameters:

- No parameters.

Examples:

```
SOUNDON
```

Result and consequences: Enables audio output.

SOUNDOFF

Availability: Local console and remote /o channel.

Accepted aliases: CSOUNDOFF.

Meaning: Disables audio output.

General syntax:

```
SOUNDOFF
```

Parameters:

- No parameters.

Examples:

```
SOUNDOFF
```

Result and consequences: Disables audio output.

INTERRUPTS

Availability: Local console and remote /o channel.

Accepted aliases: CINTERRUPTS.

Meaning: Shows all interrupts known by the computer.

General syntax:

```
INTERRUPTS
```

Parameters:

- No parameters.

Examples:

```
INTERRUPTS
```

Result and consequences: Shows all interrupts known by the computer.

INTSET

Availability: Local console and remote /o channel.

Accepted aliases: CINTSET.

Meaning: Enables or disables one interrupt or every interrupt.

General syntax:

```
INTSET ID|ALL ON|OFF
```

Parameters:

- ID or ALL: target interrupt.
- ON or OFF: requested state.

Examples:

```
INTSET ALL ON
```

Result and consequences: An unknown interrupt ID produces an error.

CHIPS

Availability: Local console and remote /o channel.

Accepted aliases: CCHIPS.

Meaning: Lists the emulated chips and their identifying attributes.

General syntax:

```
CHIPS
```

Parameters:

- No parameters.

Examples:

```
CHIPS
```

Result and consequences: Lists the emulated chips and their identifying attributes.

CRTON

Availability: Local console and remote /o channel.

Accepted aliases: CCRTON.

Meaning: Enables the CRT screen effect.

General syntax:

```
CRTON
```

Parameters:

- No parameters.

Examples:

```
CRTON
```

Result and consequences: Enables the CRT screen effect.

CRTOFF

Availability: Local console and remote /o channel.

Accepted aliases: CCRTOFF.

Meaning: Disables the CRT screen effect.

General syntax:

```
CRTOFF
```

Parameters:

- No parameters.

Examples:

```
CRTOFF
```

Result and consequences: Disables the CRT screen effect.

DATASETTE

Availability: Local console and remote /o channel.

Accepted aliases: CDATASETTE.

Meaning: Shows datasette state when a supported datasette is connected.

General syntax:

```
DATASETTE
```

Parameters:

- No parameters.

Examples:

```
DATASETTE
```

Result and consequences: Shows datasette state when a supported datasette is connected.

GRIDON

Availability: Local console and remote /o channel.

Accepted aliases: CGRIDON.

Meaning: Enables the generic screen grid overlay.

General syntax:

```
GRIDON COLOR
```

Parameters:

- COLOR: mandatory numeric grid color.

Examples:

```
GRIDON 14
```

Result and consequences: Enables the screen grid using the requested color.

GRIDOFF

Availability: Local console and remote /o channel.

Accepted aliases: CGRIDOFF.

Meaning: Disables the generic screen grid overlay.

General syntax:

```
GRIDOFF
```

Parameters:

- No parameters.

Examples:

```
GRIDOFF
```

Result and consequences: Disables the screen grid overlay.

PICTURE

Availability: Local console and remote /o channel.

Accepted aliases: CPICTURE.

Meaning: Saves a screenshot of the emulator screen.

General syntax:

```
PICTURE FILE
```

Parameters:

- FILE: output PNG, JPG or BMP filename.

Examples:

```
PICTURE vic20-screen.png
```

Result and consequences: A supported extension writes the current screen; unsupported formats fail.

SETHOOK

Availability: Local console and remote /o channel.

Accepted aliases: CSETHOOK.

Meaning: Creates an execution hook.

General syntax:

```
SETHOOK ID TYPE [PARAMS...]
```

Parameters:

- ID: unique hook identifier.
- TYPE: hook kind; currently 0 stops on memory read/write.
- For TYPE 0, PARAMS are mandatory INITIAL and optional FINAL addresses.

Examples:

```
SETHOOK 1 0 $d000 $d3ff
```

Result and consequences: Creates an execution hook.

REMOVEHOOK

Availability: Local console and remote /o channel.

Accepted aliases: CREMOVEHOOK.

Meaning: Removes a hook by identifier.

General syntax:

```
REMOVEHOOK ID
```

Parameters:

- ID: hook identifier.

Examples:

```
REMOVEHOOK 1
```

Result and consequences: If the hook does not exist, no state changes.

HOOKS

Availability: Local console and remote /o channel.

Accepted aliases: CHOOKS.

Meaning: Lists active hooks.

General syntax:

```
HOOKS
```

Parameters:

- No parameters.

Examples:

```
HOOKS
```

Result and consequences: Lists active hooks.

HOOKSHELP

Availability: Local console and remote /o channel.

Accepted aliases: CHOOKSHELP.

Meaning: Shows help for supported hook types.

General syntax:

```
HOOKSHELP
```

Parameters:

- No parameters.

Examples:

```
HOOKSHELP
```

Result and consequences: Shows help for supported hook types.

CMOVEPARAMS

Availability: Local console and remote /o channel.

Accepted aliases: No public alias.

Meaning: Builds the framework's internal MoveParametersToAnswerCommand.

General syntax:

```
CMOVEPARAMS
```

Parameters:

- No parameters are executable under the current command contract.

Examples:

```
CMOVEPARAMS
```

Result and consequences: The command returns an empty answer when invoked directly; it is an internal response helper and is not advertised in the help file.

3.2 Inherited Commodore Commands

VICII

Availability: Local console and remote /o channel.

Accepted aliases: CVICII.

Meaning: Requests VIC-II state.

General syntax:

```
VICII
```

Parameters:

- No parameters.

Examples:

```
VICII
```

Result and consequences: Accepted by the inherited builder, but the VIC-20 has no VIC-II, so the command returns no chip data.

VICIIEVENTS

Availability: Local console and remote /o channel.

Accepted aliases: None in the current internal-name registration.

Meaning: Enables or disables VIC-II event visualization.

General syntax:

```
VICIIEVENTS ON|OFF
```

Parameters:

- ON: enable event display.
- OFF: disable it.

Examples:

```
VICIIEVENTS ON
```

Result and consequences: Accepted by the inherited builder, but the VIC-20 has no VIC-II, so the command has no effect.

SID

Availability: Local console and remote /o channel.

Accepted aliases: CSID.

Meaning: Requests SID state.

General syntax:

```
SID
```

Parameters:

- No parameters.

Examples:

```
SID
```

Result and consequences: Accepted by the inherited builder, but the VIC-20 has no separate SID chip; VIC-I provides its sound generation, so no SID data is returned.

SIDW

Availability: Local console and remote /o channel.

Accepted aliases: CSIDW.

Meaning: Changes the active SID sound wrapper.

General syntax:

```
SIDW LIBID
```

Parameters:

- 0: reSID wrapper.
- 1: ICF/simple wrapper.

Examples:

```
SIDW 0  
SIDW 1
```

Result and consequences: Accepted by the inherited builder, but the VIC-20 has no SID chip, so the sound implementation is unchanged.

VICI

Availability: Local console and remote /o channel.

Accepted aliases: CVICI.

Meaning: Shows VIC-I video, raster and sound state.

General syntax:

```
VICI
```

Parameters:

- No parameters.

Examples:

```
VICI
```

Result and consequences: Returns the VIC-I state used by the VIC-20.

TED

Availability: Local console and remote /o channel.

Accepted aliases: CTED.

Meaning: Requests TED state.

General syntax:

```
TED
```

Parameters:

- No parameters.

Examples:

```
TED
```

Result and consequences: Accepted by the inherited builder, but the VIC-20 has no TED chip, so the command returns no chip data.

TEDEVENTS

Availability: Local console and remote /o channel.

Accepted aliases: None in the current internal-name registration.

Meaning: Enables or disables TED event visualization.

General syntax:

```
TEDEVENTS ON|OFF
```

Parameters:

- ON or OFF: requested event-display state.

Examples:

```
TEDEVENTS ON
```

Result and consequences: Accepted by the inherited builder, but the VIC-20 has no TED chip, so the command has no effect.

VIA

Availability: Local console and remote /o channel.

Accepted aliases: CVIA.

Meaning: Requests the generic Commodore VIA state.

General syntax:

```
VIA
```

Parameters:

- No parameters.

Examples:

```
VIA
```

Result and consequences: The VIC-20 registers its controllers as VIA1 and VIA2 rather than the generic VIA identifier; use VIA1 or VIA2 for deterministic output.

CIA

Availability: Local console and remote /o channel.

Accepted aliases: CCIA.

Meaning: Requests generic Commodore CIA state.

General syntax:

```
CIA
```

Parameters:

- No parameters.

Examples:

```
CIA
```

Result and consequences: Accepted by the inherited builder, but the VIC-20 uses VIA1 and VIA2 and has no CIA, so no CIA data is returned.

3.3 VIC20-Specific Commands

VIA1

Availability: Local console and remote /o channel.

Accepted aliases: CVIA1.

Meaning: Shows the complete state of VIA #1.

General syntax:

```
VIA1
```

Parameters:

- No parameters.

Examples:

```
VIA1
```

Result and consequences: Returns the VIC-20 VIA #1 register, timer, port and interrupt state.

VIA2

Availability: Local console and remote /o channel.

Accepted aliases: CVIA2.

Meaning: Shows the complete state of VIA #2.

General syntax:

```
VIA2
```

Parameters:

- No parameters.

Examples:

```
VIA2
```

Result and consequences: Returns the VIC-20 VIA #2 register, timer, port and interrupt state.

SCREENDUMP

Availability: Local console and remote /o channel.

Accepted aliases: CSCREENDUMP.

Meaning: Returns the current text-screen memory in hexadecimal.

General syntax:

```
SCREENDUMP
```

Parameters:

- No parameters.

Examples:

```
SCREENDUMP
```

Result and consequences: Returns the VIC-I screen-memory snapshot visible to the active CPU.

COLORDUMP

Availability: Local console and remote /o channel.

Accepted aliases: CCOLORDUMP.

Meaning: Returns the current color memory in hexadecimal.

General syntax:

```
COLORDUMP
```

Parameters:

- No parameters.

Examples:

```
COLORDUMP
```

Result and consequences: Returns the VIC-I color-memory snapshot visible to the active CPU.

CHARSDRAW

Availability: Local console and remote /o channel.

Accepted aliases: CCHARSDRAW.

Meaning: Renders textual drawings of all or selected character glyphs.

General syntax:

```
CHARSDRAW [0..255 ...]
```

Parameters:

- Character codes: optional values from 0 through 255; duplicates are coalesced. With none, all characters are rendered.

Examples:

```
CHARSDRAW
```

```
CHARSDRAW 1 2 65
```

Result and consequences: Returns textual drawings generated from VIC-I character data.

LIGHTPEN

Availability: Local console and remote /o channel.

Accepted aliases: CLIGHTPEN.

Meaning: Enables or disables mouse-driven light-pen emulation.

General syntax:

```
LIGHTPEN ON|OFF
```

Parameters:

- ON: enable.
- OFF: disable.

Examples:

```
LIGHTPEN ON  
LIGHTPEN OFF
```

Result and consequences: The mouse moves the VIC-I light pen and its left button presses it. This implementation can coexist with joysticks.

3.4 Local-Console-Only Commands

LOADPRG

Availability: Local console only.

Accepted aliases: None.

Meaning: Loads, assembles and installs an assembler source through the emulator object.

General syntax:

```
LOADPRG FILE
```

Parameters:

- FILE: assembler source path.

Examples:

```
LOADPRG demo.asm
```

Result and consequences: Compilation or loading errors are reported by the local console.

LOADBINARY

Availability: Local console only.

Accepted aliases: None.

Meaning: Loads a binary containing its destination address in the file header.

General syntax:

```
LOADBINARY FILE
```

Parameters:

- FILE: header-addressed binary path.

Examples:

```
LOADBINARY demo.prg
```

Result and consequences: Loads a binary containing its destination address in the file header.

LOADBLOCKS

Availability: Local console only.

Accepted aliases: None.

Meaning: Loads a block-structured memory file.

General syntax:

```
LOADBLOCKS FILE
```

Parameters:

- FILE: block file path.

Examples:

```
LOADBLOCKS state.blocks
```

Result and consequences: Loads a block-structured memory file.

DECOMPILE

Availability: Local console only.

Accepted aliases: None.

Meaning: Disassembles a memory region.

General syntax:

```
DECOMPILE ADDRESS BYTES
```

Parameters:

- ADDRESS: first address.
- BYTES: approximate byte count; output rounds to complete instructions.

Examples:

```
DECOMPILE $080d 64
```

Result and consequences: Disassembles a memory region.

POSSIBLEPERS

Availability: Local console only.

Accepted aliases: None.

Meaning: Lists peripheral IDs and required attributes supported by the active I/O peripheral builder.

General syntax:

```
POSSIBLEPERS
```

Parameters:

- No parameters.

Examples:

```
POSSIBLEPERS
```

Result and consequences: Lists peripheral IDs and required attributes supported by the active I/O peripheral builder.

CONNECTPER

Availability: Local console only.

Accepted aliases: None.

Meaning: Builds and connects a peripheral to the first compatible device.

General syntax:

```
CONNECTPER ID [ATTRS...]
```

Parameters:

- ID: peripheral-emulation ID from POSSIBLEPERS.
- ATTRS: optional device-specific positional attributes, such as d:8 for a 1541 or d:4 f:printer.ps p:MPS801-PS for a printer.

Examples:

```
CONNECTPER 100  
CONNECTPER 215 d:8
```

Result and consequences: Connection succeeds only when the ID and attributes are valid and a compatible device is available.

DISCONNECTPERS

Availability: Local console only.

Accepted aliases: None.

Meaning: Disconnects one or more peripherals.

General syntax:

```
DISCONNECTPERS ID1 [ID2 ...]
```

Parameters:

- IDn: connected peripheral IDs.

Examples:

```
DISCONNECTPERS 100 101
```

Result and consequences: Disconnects one or more peripherals.

LOADPERDATA

Availability: Local console only.

Accepted aliases: None.

Meaning: Loads media/data into a connected peripheral.

General syntax:

```
LOADPERDATA ID FILE [DEBUGFILE]
```

Parameters:

- ID: connected peripheral ID.
- FILE: supported media/data file.
- DEBUGFILE: optional file used to start format-related debugging.

Examples:

```
LOADPERDATA 215 disk.d64
```

Result and consequences: The operation fails if the peripheral does not support the supplied format.

EMPTYPERDATA

Availability: Local console only.

Accepted aliases: None.

Meaning: Creates empty media/data in a peripheral.

General syntax:

```
EMPTYPERDATA ID FILE
```

Parameters:

- ID: connected peripheral ID.
- FILE: path whose extension selects a supported creatable format.

Examples:

```
EMPTYPERDATA 215 blank.d64
```

Result and consequences: Only peripherals and formats that implement empty-media creation can succeed.

SAVEPERDATA

Availability: Local console only.

Accepted aliases: None.

Meaning: Saves a peripheral's current media/data.

General syntax:

```
SAVEPERDATA ID [FILE]
```

Parameters:

- ID: connected peripheral ID.
- FILE: optional output path; may be omitted when LOADPERDATA or EMPTYPERDATA established the current filename.

Examples:

```
SAVEPERDATA 215  
SAVEPERDATA 215 copy.d64
```

Result and consequences: Saves a peripheral's current media/data.

CLEARPERDATA

Availability: Local console only.

Accepted aliases: None.

Meaning: Clears all data currently loaded into peripherals.

General syntax:

```
CLEARPERDATA
```

Parameters:

- No parameters.

Examples:

```
CLEARPERDATA
```

Result and consequences: Commonly used to force a subsequent program/media reload.

IGNACIO CEA FORNIES

Appendices

Appendix A. Command Availability Summary

The table distinguishes the interactive console from the command channel of the non-console executable. Remote availability assumes VIC20Emulator.exe was started with /o and that the client is connected.

Command family	VIC20EmulatorC local console	VIC20Emulator remote /o
Standard builder	Yes	Yes
Inherited Commodore builder	Yes	Yes
VIC20 builder	Yes	Yes
LocalConsole emulator/media operations	Yes	No

Appendix B. Source-of-Truth Maintenance Rule

This file is a maintained repository artifact. Every change that adds, removes, renames or changes a VIC-20 startup option, a command-builder registration, a LocalConsole-only command, command syntax, parameter meaning, command consequence or user-visible formatted answer must include a review and, when applicable, an update of docs/VIC20Emulator_UserGuide.docx. The corresponding canonical .fmt source must be updated whenever the command's InfoStructure contract changes.

Audit the complete chain, not only the edited class: VIC20Emulator entry point and VIC20::Emulator for startup options; LocalConsole for local-only commands; and VIC20::CommandBuilder -> COMMODORE::CommandBuilder -> MCHEmul::StandardCommandBuilder for builder commands.

Illustrations

NO ILLUSTRATIONS ARE INCLUDED IN THIS EDITION.

IGNACIO CEA FORNIÉS

Equations

NO EQUATIONS ARE INCLUDED IN THIS EDITION.

IGNACIO CEA FORNIÉS